THE INTERNAL HOM EVALUATION MAP IS AN ISOMORPHISM UNDER BOUNDEDNESS AND FINITE INJECTIVE DIMENSION HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G4I (sources/stacks-duality/pinned-original-source.tex line 415).

Allowed standard prerequisites (editorial).

- The evaluation map between internal Homs. Cohomology, Lemma labelled cohomology-lemma-internal-hom-evaluate, invoked in the statement. Supplies the arrow whose bijectivity is asserted.
- Comparison of the affine and the scheme level descriptions. Same chapter, Lemma labelled lemma-affine-duality, cited at proof lines 454-458 and 464-466, and Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-internal-hom, cited at proof lines 456-460. Shows both sides commute with the functor from D(A) to D_QCoh(O_X).
- Boundedness of the internal Hom of coherent complexes. Derived Categories of Schemes, Lemma labelled perfect-lemma-coherent-internal-hom, cited at proof lines 450-454, and Dualizing Complexes, Lemma labelled dualizing-lemma-finite-ext-into-bounded-injective, cited at proof lines 466-468. Gives RSheafHom(K, L) in D^+_Coh in case (1) and in D_Coh in case (2).
- The algebraic evaluation isomorphisms. More on Algebra, Lemmas labelled more-algebra-lemma-internal-hom-evaluate-isomorphism parts (1) and (2), cited at proof lines 460-462 and 468-469. Supply the isomorphism after the reduction to affines.

Source reference. Original lemma 0A8U: sources/stacks-chow/tags.txt, source lines 12927–12927. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is

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Source reference. Original lemma 06Z0: sources/stacks-chow/tags.txt, source lines 8904–8904. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 440 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0D0C: sources/stacks-chow/tags.txt, source lines 16307–16307. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 448 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A88: sources/stacks-duality/pinned-original-source.tex, source lines 366–366. Same-chapter pointer into the author environment lemma at source line 366; invoked at proof line 451 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 453 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A68: sources/stacks-chow/tags.txt, source lines 12836–12836. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 458 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4H: sources/stacks-chow/tags.txt, source lines 20127–20127. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 465 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A69: sources/stacks-chow/tags.txt, source lines 12837–12837. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 468 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a Noetherian scheme. Let $K, L, M \in D_{QCoh}(\mathcal{O}_X)$. Then the map*

$$R\mathcal{H}om(L, M) \otimes^{\mathbf{L}}_{\mathcal{O}_X} K \longrightarrow R\mathcal{H}om(R\mathcal{H}om(K, L), M)$$

of Cohomology, Lemma 0A8U is an isomorphism in the following two cases

1. *$K \in D^-_{Coh}(\mathcal{O}_X)$, $L \in D^+_{Coh}(\mathcal{O}_X)$, and M affine locally has finite injective dimension (see proof), or*
2. *K and L are in $D_{Coh}(\mathcal{O}_X)$, the object $R\mathcal{H}om(L, M)$ has finite tor dimension, and L and M affine locally have finite injective dimension (in particular L and M are bounded).*

Proof. Proof of (1). We say M has affine locally finite injective dimension if X has an open covering by affines $U = \mathrm{Spec}(A)$ such that the object of $D(A)$ corresponding to $M|_U$ (Derived Categories of Schemes, Lemma 06Z0) has finite injective dimension[1]. To prove the lemma we may replace X by U, i.e., we may assume $X = \mathrm{Spec}(A)$ for some Noetherian ring A. Observe that $R\mathcal{H}om(K, L)$ is in $D^+_{Coh}(\mathcal{O}_X)$ by Derived Categories of Schemes, Lemma 0D0C. Moreover, the formation of the left and right hand side of the arrow commutes with the functor $D(A) \to D_{QCoh}(\mathcal{O}_X)$ by Lemma 0A88 and Derived Categories of Schemes, Lemma 0A6H (to be sure this uses the assumptions on K, L, M and what we just proved about $R\mathcal{H}om(K, L)$). Then finally the arrow is an isomorphism by More on Algebra, Lemmas 0A68 part (2).

Proof of (2). We argue as above. A small change is that here we get $R\mathcal{H}om(K, L)$ in $D_{Coh}(\mathcal{O}_X)$ because affine locally (which is allowable by Lemma 0A88) we may appeal to Dualizing Complexes, Lemma 0G4H. Then we finally conclude by More on Algebra, Lemma 0A69. □

[1]This condition is independent of the choice of the affine open cover of the Noetherian scheme X. Details omitted.